

R for bioinformatics, data wrangler practices

HUST Bioinformatics course series

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Section 1

section 1: TOC

前情提要

pipe

- `%>%`
- `%$%`
- `%T>%`
- `%<>%`

tidyr

- `pivot_longer()`
- `pivot_wider()`
- `unite()`
- `separate()`

前情提要, cont.

dplyr

- `select()`
- `filter()`
- `mutate()`
- `summarise()`
- `arrange()`
- `group_by()` ...

本次提要

- ① 3 个生信任务的 R 解决方案
- ② factors 的更多应用 (forcats)

Section 2

section 2: contents

生信任务 1 : network analysis and visualisation (dplyr & some plot packages)

protein-protein interaction data

why PPI is important?

- ① most of the time, protein functions together with other proteins (interactions)
- ② interacting partners tend to have similar functions (guilty by association, can be used in gene annotation)

STRING database

- ① contains >2 billion interactions for 24.6 million proteins in 5090 organisms (as of May 2021; ver 11.0b)
- ② contains:
 - physical interaction
 - genetic interactions
 - gene co-occurrence (text-mining)
 - transfers through orthologous relationships

STRING is one of the most cited resources

STRING v11: protein–protein association networks with increased coverage, supporting functional discovery in genome-wide experimental datasets

D Szklarczyk, AL Gable, D Lyon, A Junge, S Wyder, J Huerta-Cepas, ...
Nucleic acids research 47 (D1), D607-D613

3973

2019

screenshot took in May 2021

more tools at Peer Bork's group: https://scholar.google.com/citations?hl=en&user=M6Etr6oAAAAJ&view_op=list_works&sortby=pubdate

a typical STRING plot

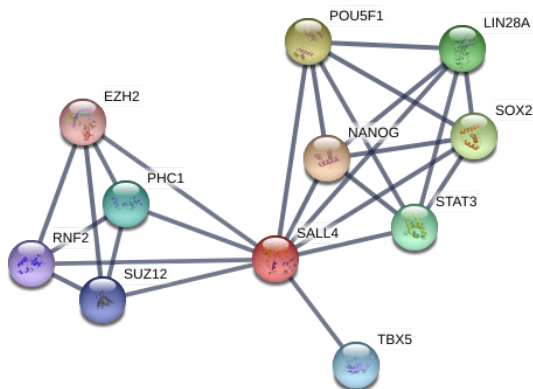


Figure 1: interacting partners of SALL4

note to previous plot

note this plot was generated using the following parameters:

- use SALL4 in human as query
- show only the top ten connectivity partners
- only connectivity score ≥ 900 (or 0.9) were shown

tasks of task 1

- ① get human PPI data
- ② limit the interactions to those with scores ≥ 900 (or 0.9)
- ③ find SALL4 and its top ten interaction partners
- ④ visualize the PPI network
- ⑤ calculate connectivities of sall4 and its top ten partners in the dataset

问题

生物学意义何在??

download human ppi data from STRING

go to : <https://string-db.org/cgi/download.pl>

FILES TOO LARGE?

Some of the files below can be made smaller (prior to download), by restricting the data to one organism of interest. Choose an organism here:

Homo sapiens

▼

UPDATE

INTERACTION DATA

download

File	Description	Access
9606.protein.links.v11.0.txt.gz (71.2 Mb)	protein network data (scored links between proteins)	
9606.protein.links.detailed.v11.0.txt.gz (110.1 Mb)	protein network data (incl. subscores per channel)	
9606.protein.links.full.v11.0.txt.gz (127.6 Mb)	protein network data (incl. distinction: direct vs. interologs)	
9606.protein.actions.v11.0.txt.gz (14.4 Mb)	interaction types for protein links	

Figure 2: Download human PPI data from STRING

load and load the human PPI data

```
library(tidyverse);
```

```
## read_csv 也能处理压缩文件 !!!
```

```
ppi <- read_delim( file = "data/talk06/ppi900.txt.gz", col_names = T,
                  delim = "\t", quote = "" );
```

```
## 查看一下数据 --
```

```
ppi %>% filter( gene1 == "SALL4" ) %>% do( head(., n = 10 ) );
```

```
# A tibble: 10 x 3
```

	gene1	gene2	score
	<chr>	<chr>	<dbl>
1	SALL4	POLR2E	900
2	SALL4	POLR2C	900
3	SALL4	POLR2I	900
4	SALL4	NANOG	992
5	SALL4	SALL1	923
6	SALL4	ZSCAN10	912
7	SALL4	LIN28A	957
8	SALL4	POU5F1	986
9	SALL4	SMAD2	906
10	SALL4	EPAS1	900



start to process the data

```
## get top 10 interacting partners of SALL4 by interaction score ...
toppart <- ppi %>% filter( gene1 == "SALL4" ) %>%
  arrange( desc( score ) ) %>% slice( 1:10 );

## get the interaction network consisting the top genes --
genes <- unique( c( "SALL4", toppart$gene2 ) );
netdata <- ppi %>% filter( gene1 %in% genes & gene2 %in% genes );
nrow(netdata);
```

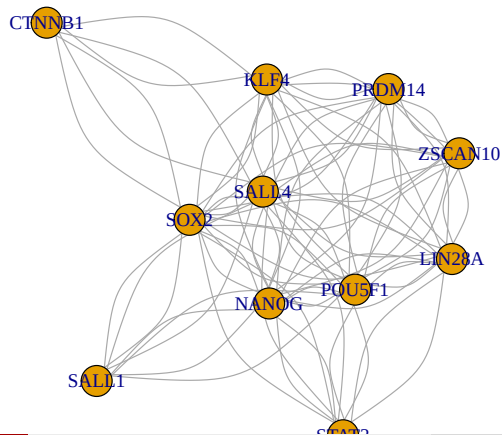
```
[1] 80
```


load the igraph package

```
## -- to make sure the installation will only run once ...  
if (!require("igraph")){  
  chooseCRANmirror();  
  install.packages("igraph");  
}  
  
library( igraph );
```

visualise the network ...

```
netnet <- graph_from_data_frame( netdata, directed = FALSE );  
plot(netnet);
```



图的问题

- very ugly
- two lines (redundancy) between every two nodes

redundancy among data

```
netdata %>% filter( gene1 %in% c("SALL4", "NANOG") & gene2 %in% c("SALL4", "NANOG") );
```

```
# A tibble: 2 x 3  
  gene1 gene2 score  
  <chr> <chr> <dbl>  
1 NANOG SALL4   992  
2 SALL4 NANOG   992
```

how to remove redundancy?

```
## create a new column, sort the two gene names, and concatenate them ...
testdata <-
  netdata %>% filter( gene1 %in% c("SALL4", "NANOG") & gene2 %in% c("SALL4", "NANOG") ) %>%
  mutate( group =
    if_else( gene1 > gene2,
      str_c( gene1, gene2, sep = "-" ),
      str_c( gene2, gene1, sep = "-" ) ) );

testdata;
```

```
# A tibble: 2 x 4
  gene1 gene2 score group
<chr> <chr> <dbl> <chr>
1 NANOG SALL4   992 SALL4-NANOG
2 SALL4 NANOG   992 SALL4-NANOG
```

Note `str_c` is from the `stringr` package!!



remove redundancy!

```
testdata %>% group_by( group ) %>% slice( 1 );
```

```
# A tibble: 1 x 4  
# Groups:   group [1]  
  gene1 gene2 score group  
  <chr> <chr> <dbl> <chr>  
1 NANOG SALL4   992 SALL4-NANOG
```

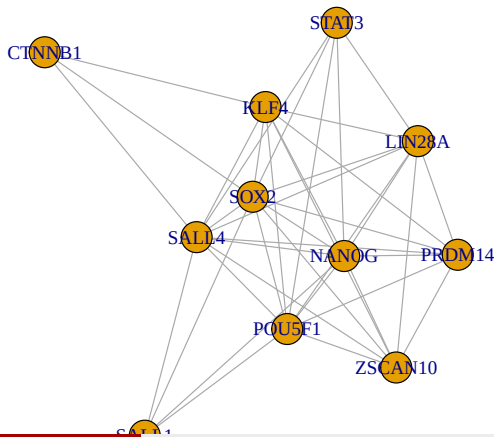
remove redundancy, cont.

```
netdata.nr <-  
  netdata %>%  
  mutate( group =  
    if_else( gene1 > gene2,  
            str_c( gene1, gene2, sep = "-" ),  
            str_c( gene2, gene1, sep = "-" ) ) ) %>%  
  group_by( group ) %>% slice( 1 );  
  
nrow(netdata.nr);
```

```
[1] 40
```

plot the non-redundant data

```
netnet.nr <- graph_from_data_frame( netdata.nr, directed = FALSE );  
plot(netnet.nr);
```



redundant 的数据可以用来计算 degree

```
net.stats <-
  netdata %>% group_by( gene1 ) %>% summarise( degree = n() ) %>%
  arrange( desc (degree) );
net.stats;
```

A tibble: 11 x 2

	gene1 <chr>	degree <int>
1	SALL4	10
2	SOX2	10
3	NANOG	9
4	POU5F1	9
5	KLF4	8
6	LIN28A	8
7	PRDM14	7
8	ZSCAN10	7
9	STAT3	5
10	SALL1	4
11	CTNNB1	3

继续美化 network

```
## node 大小由 degree 决定
## 查看网络中基因名存储的顺序
V(netnet.nr)$name;
```

```
[1] "KLF4"      "LIN28A"    "POU5F1"    "PRDM14"    "SALL1"      "CTNNB1"    "NANOG"
[8] "SOX2"      "STAT3"     "ZSCAN10"   "SALL4"
```

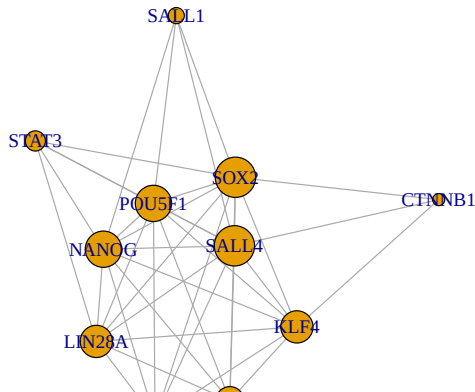
```
## 获取它们相对应的 degree ...
net.stats[match( V(netnet.nr)$name , net.stats$gene1 ),  ];
```

```
# A tibble: 11 x 2
```

	gene1	degree
	<chr>	<int>
1	KLF4	8
2	LIN28A	8
3	POU5F1	9
4	PRDM14	7
5	SALL1	4
6	CTNNB1	3
7	NANOG	9
8	SOX2	10
9	STAT3	5

继续美化 network, cont.

```
vertex_attr(netnet.nr, "size") <-
  net.stats$degree[match( V(netnet.nr)$name , net.stats$gene1 ) ] * 2;
plot( netnet.nr );
```



更多美化

详见:

- 1 igraph introduction
- 2 a comprehensive network visualization tutorial in R

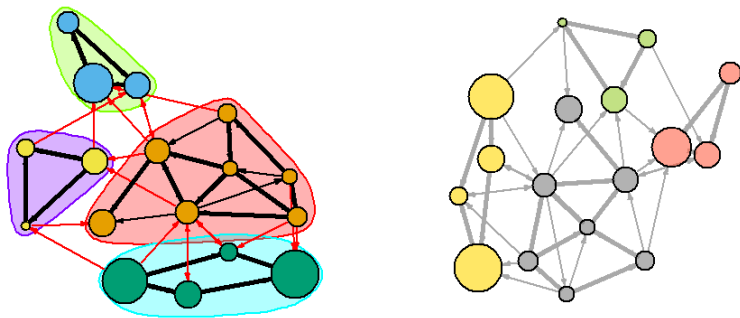


Figure 3: Example final outcomes

other network visualisation packages

- ① ggnet
- ② interactive networkD3 R package
- ③ plotly.js
- ④ D3

生信任务 2：宏基因组数据展示的小应用 (forcats)

Relative abundances

Species

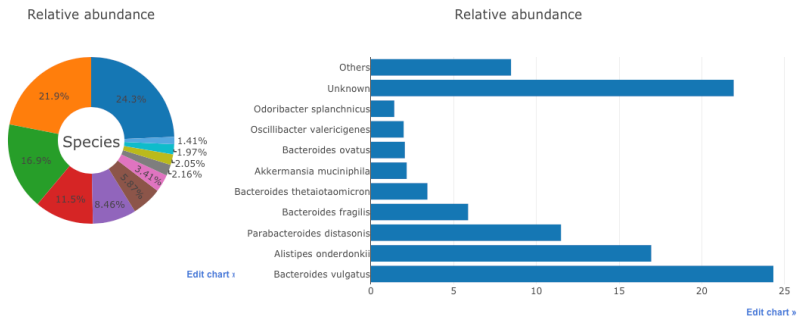


Figure 4: 肠道物种丰度示意图

Find more at the GMrepo database.

什么是宏基因组?

Metagenomics is the study of genetic material recovered directly from environmental samples.

research contents

- mostly prokaryotes
- unicellular eukaryotes
- viruses

techniques

- 16S (universally conserved gene in prokaryotes)
- whole genome sequencing (WGS or metagenomics)

what can metagenomics do?

- identify new species
- reveal species kinds and abundances
- relate species changes to human health and disease

biomes

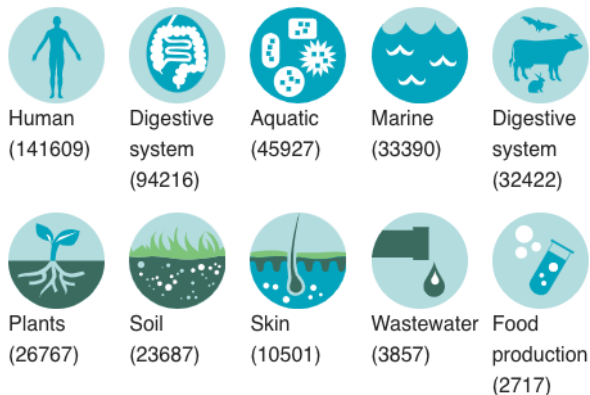


Figure 5: Biomes

Screenshot taken from the EBI MGnify database on Aug 4, 2021;

enviromental microbiome

- soil
- ocean



Figure 6: The tara oceans expedition

host associated microbiomes

- human body sites
- animals & plants



Figure 7: NIH human microbiome project

why human gut microbiota is important?

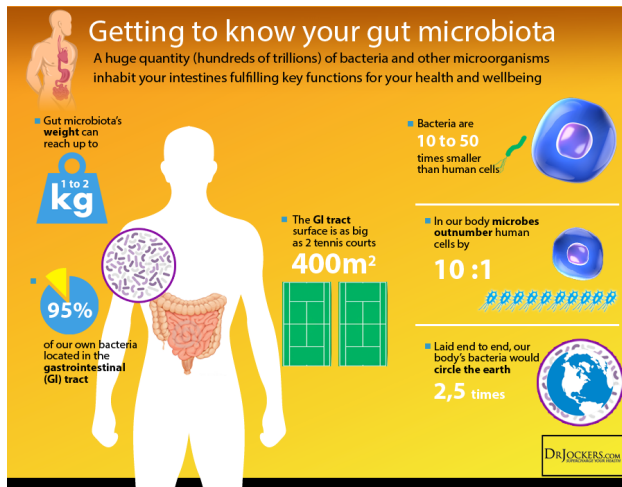


Figure 8: human gut microbiota

tasks of human gut microbiota analysis

- identify new species
- find good, bad and commensal microbes
- link microbial variations to human health
- mechanisms
- modulation, intervention and regulation

typical human gut microbiome data

Species abundances

```
abu <-  
  read_delim(  
    file = "data/talk06/relative_abundance_for_RUN_ERR1072629_taxonlevel_species.txt",  
    delim = "\t", quote = "", comment = "#");  
  
nrow(abu);
```

```
[1] 122
```

Species abundances, cont.

```
abu %>% arrange( desc( relative_abundance ) ) %>% do( head(., n = 10) );
```

```
# A tibble: 10 x 3
```

	ncbi_taxon_id	relative_abundance	scientific_name
	<dbl>	<dbl>	<chr>
1	821	24.3	Bacteroides vulgatus
2	-1	21.9	Unknown
3	328813	16.9	Alistipes onderdonkii
4	823	11.5	Parabacteroides distasonis
5	817	5.87	Bacteroides fragilis
6	818	3.41	Bacteroides thetaiotaomicron
7	239935	2.16	Akkermansia muciniphila
8	28116	2.05	Bacteroides ovatus
9	351091	1.97	Oscillibacter valericigenes
10	28118	1.41	Odoribacter splanchnicus

相对丰度作图要求

- ① 按丰度从高到低排序
- ② 只取前 10 个 species (保留 10 行)
- ③ 将后面的丰度累加在一起, 汇总为"Others" 分类

数据处理

```
library( tidytidbits );
abu.dat <-
  abu %>% arrange( desc( relative_abundance ) ) %>%
    lump_rows( scientific_name, relative_abundance, n = 10, other_level = "Others" );

head(abu.dat, n = 11);
```

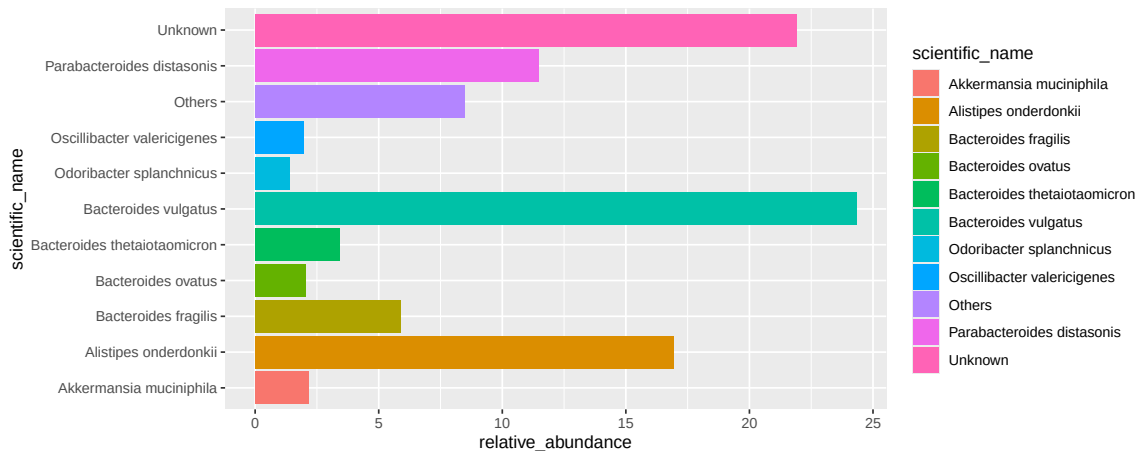
A tibble: 11 x 3

	ncbi_taxon_id	relative_abundance	scientific_name
	<dbl>	<dbl>	<chr>
1	821	24.3	Bacteroides vulgatus
2	-1	21.9	Unknown
3	328813	16.9	Alistipes onderdonkii
4	823	11.5	Parabacteroides distasonis
5	817	5.87	Bacteroides fragilis
6	818	3.41	Bacteroides thetaiotaomicron
7	239935	2.16	Akkermansia muciniphila
8	28116	2.05	Bacteroides ovatus
9	351091	1.97	Oscillibacter valericigenes
10	28118	1.41	Odoribacter splanchnicus
11	31199992	8.46	Others



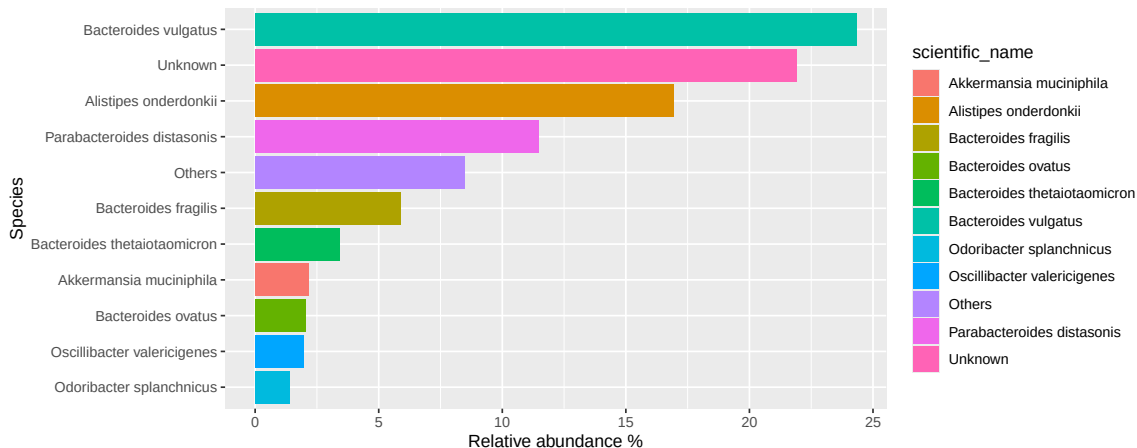
尝试作图

```
ggplot(abu.dat, aes(x = scientific_name, y = relative_abundance, fill = scientific_name ) ) +
  geom_bar( stat = "identity" ) +
  coord_flip()
```



调整排列顺序

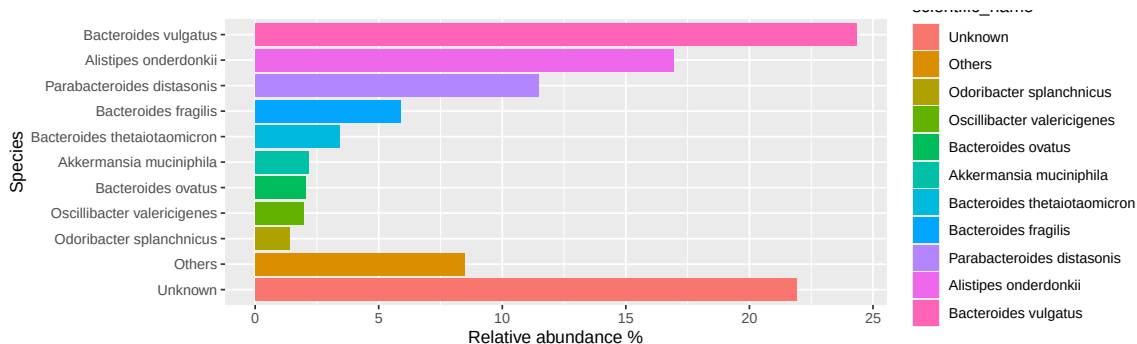
```
## 用 forcat 包的 fct_reorder 函数; 注意它 3 个参数的意义 !!!
ggplot(abu.dat, aes(x = fct_reorder( scientific_name, relative_abundance, .desc = F),
                                y = relative_abundance, fill = scientific_name )) +
  geom_bar( stat = "identity" ) +
  coord_flip() + xlab("Species") + ylab( "Relative abundance %" )
```



把 Others 和 Unknown 放在最后

注意 `fct_reorder` 的用法:

```
abu.dat$scientific_name <-
  fct_relevel( fct_reorder( abu.dat$scientific_name, abu.dat$relative_abundance, .desc = F),
              "Unknown", "Others" );
ggplot(abu.dat, aes(x = scientific_name,
                    y = relative_abundance, fill = scientific_name ) ) +
  geom_bar( stat = "identity" ) +
  coord_flip() + xlab("Species") + ylab( "Relative abundance %" )
```



更多 forcats 的应用 ...

see here: <https://cran.r-project.org/web/packages/forcats/vignettes/forcats.html>

更多应用将会在 作图 (ggplot2) 时讲到。

生信任务 3：整合基因表达、甲基化、突变数据 (dplyr::join)

在生信分析中，常需要将多个来源的数据整合在一个表格中，以方便后续分析。

```
meth <- read_delim( file = "data/talk06/methylation_data.txt.gz",
                    delim = "\t", quote = "", col_names = T);

head(meth, n = 3);
```

```
# A tibble: 3 x 3
  gene   site   methylation_score
  <chr> <chr>             <dbl>
1 A1BG  1stExon           0.799
2 A1BG  5UTR              0.799
3 A1BG  Body              0.811
```

问题 什么是甲基化??

表达数据

```
expr <- read_delim( file = "data/talk06/expression_data.txt.gz",  
                    delim = "\t", quote = "", col_names = T );  
  
head( expr, n = 5);
```

```
# A tibble: 5 x 2  
  gene      rkpm  
  <chr>   <dbl>  
1 5S_RRNA     0  
2 7SK         0  
3 A1BG        1  
4 A1BGAS1     4  
5 A1CF        0
```

整合后的结果应该是什么??

gene	TSS200	TSS1500	UTR	body	expression
gene1	0.1	0.2	NA	0.8	100
gene2	0.12	0.32	NA	0.9	18

Figure 9: 整合后的结果示意

第一种方法, 使用 `pivot_wider`

思路: 先合并, 再 `pivot_wider`

① 准备数据

```
## 用 `bind_rows` 合并两个 tibble 时, 列名需要一致
meth2 <- meth %>% select( gene, group=site, value=methylation_score );
head(meth2, n=2);
```

```
# A tibble: 2 x 3
  gene  group  value
  <chr> <chr>  <dbl>
1 A1BG  1stExon 0.799
2 A1BG  5UTR    0.799
```

```
expr2 <- expr %>% mutate( group = "rkpm" ) %>%
  select( gene, group, value=rkpm ) %>% group_by( gene ) %>% slice( 1 );
```

第一种方法，使用 `pivot_wider`，准备数据，cont.

```
## 查看数据;  
head(expr2, n=2);
```

```
# A tibble: 2 x 3  
# Groups:   gene [2]  
  gene    group value  
  <chr>  <chr> <dbl>  
1 5S_RRNA rkpm      0  
2 7SK     rkpm      0
```

注: gene name 与 group 组合必须是唯一的。即: 基因 A 只能有一个表达量值。

合并 & pivot_wider

2 合并

3 prvot_wider

```
library(kableExtra);
comb <- bind_rows( meth2, expr2 );

comb.wide <- comb %>% pivot_wider( names_from = "group", values_from = "value" );

kbl( head(comb.wide) );
```

gene	1stExon	5UTR	Body	TSS1500	TSS200	3UTR	rkpm
A1BG	0.7991930	0.799193	0.8112005	0.852340	0.849514	NA	1
A1CF	0.0945684	0.118445	0.0079776	0.109668	NA	NA	0
A2M	0.8993690	NA	0.7161885	0.653861	NA	0.776134	1
A2ML1	0.8590680	0.859068	0.6783700	0.440733	NA	0.469400	0
A4GALT	NA	0.844301	0.8209765	0.889742	0.512742	0.813896	0
A4GNT	0.6801970	0.680197	0.5378910	0.401741	NA	NA	0

方法二: 使用 join ...

首先对 methylation 数据进行处理

```
meth3 <-  
  meth %>% pivot_wider( names_from = "site", values_from = "methylation_score" );  
  
kbl( head(meth3) );
```

gene	1stExon	5UTR	Body	TSS1500	TSS200	3UTR
A1BG	0.7991930	0.799193	0.8112005	0.852340	0.849514	NA
A1CF	0.0945684	0.118445	0.0079776	0.109668	NA	NA
A2M	0.8993690	NA	0.7161885	0.653861	NA	0.776134
A2ML1	0.8590680	0.859068	0.6783700	0.440733	NA	0.469400
A4GALT	NA	0.844301	0.8209765	0.889742	0.512742	0.813896
A4GNT	0.6801970	0.680197	0.5378910	0.401741	NA	NA

dplyr::join

```
comb2 <- left_join( meth3, expr2, by = "gene" ) %>% select( -group );
kbl( head(comb2) );
```

gene	1stExon	5UTR	Body	TSS1500	TSS200	3UTR	value
A1BG	0.7991930	0.799193	0.8112005	0.852340	0.849514	NA	1
A1CF	0.0945684	0.118445	0.0079776	0.109668	NA	NA	0
A2M	0.8993690	NA	0.7161885	0.653861	NA	0.776134	1
A2ML1	0.8590680	0.859068	0.6783700	0.440733	NA	0.469400	0
A4GALT	NA	0.844301	0.8209765	0.889742	0.512742	0.813896	0
A4GNT	0.6801970	0.680197	0.5378910	0.401741	NA	NA	0

注意 left_join 的语法

join 详解

- `left_join()`: return all rows from x, and all columns from x and y. Rows in x with no match in y will have NA values in the new columns. If there are multiple matches between x and y, all combinations of the matches are returned.
- `inner_join()`
- `right_join()`
- `full_join()`

更多请见: <https://dplyr.tidyverse.org/reference/join.html>

Section 3

section 3: Exercise & home work

小结

今次提要

- ① 3 个生信任务的 R 解决方案
- ② factors 的更多应用 (forcats)

下次预告

- Strings and regular expression

important

- all codes are available at Github:
<https://github.com/evolgeniusteam/R-for-bioinformatics>

练习 & 作业

- Exercises and homework 目录下 talk06-homework.Rmd 文件;
- 完成时间: 见钉群的要求